
Sonic Signatures

Frequency-Domain Audio Identification

EE200 Course Project Report – Q3A

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Live Application

<https://sonicsignatures-suryapingali.streamlit.app/>

Source Code

http://github.com/surya-pingali/Sonic_Signatures

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1 Spectrograms and Resolution Trade-offs

The foundational step in the audio identification pipeline requires transforming the one-dimensional discrete audio signal into a two-dimensional time-frequency representation. This is achieved using the Short-Time Fourier Transform (STFT), which applies a sliding window function to the signal.

The choice of window length ('nperseg') dictates the resolution of the resulting spectrogram, governed by the Gabor limit (uncertainty principle) in signal processing. A signal cannot be simultaneously localized with infinite precision in both time and frequency domains:

$$\Delta t \Delta f \geq \frac{1}{4\pi} \quad (1)$$

As demonstrated in Figure 1, modifying the window length forces a strict trade-off. A short window length ($N = 256$) provides excellent temporal resolution, precisely isolating the onset of transient sounds like percussion. However, it results in poor frequency resolution, visible as vertical smearing across the frequency bins. Conversely, a long window length ($N = 4096$) provides high-definition frequency isolation, separating closely spaced harmonics, but smears the temporal boundaries horizontally, making precise timing extraction difficult.

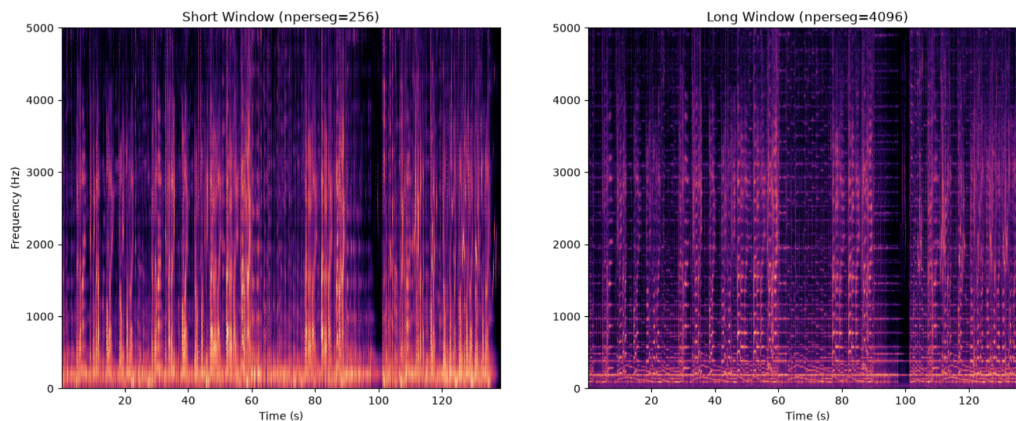


Figure 1: Spectrograms showing the trade-off between time and frequency resolution. A short window (left) gives exact timing but blurry frequencies, while a long window (right) gives clear frequencies but blurry timing.

2 Hashing Efficiency: Single Peaks vs. Paired Hashes

After extracting the maximum amplitude peaks from the spectrogram to form a feature constellation, the system must generate matching tokens. Evaluating the system using isolated frequency peaks reveals a severe structural limitation due to combinatorial explosion.

An isolated frequency peak f_1 lacks uniqueness. Common frequencies (e.g., standard musical notes) appear millions of times across a multi-song database. Querying a database using single peaks generates an overwhelming number of false-positive matches (collisions), as shown by the high noise floor in Figure 2(a).

To enforce specificity, the algorithm implements paired hashing. A hash is constructed as a 3-tuple comprising two frequencies and the temporal distance between them:

$$H = (f_1, f_2, \Delta t) \quad (2)$$

By tethering two distinct frequencies over a specific time interval, the resulting token becomes highly unique. As demonstrated in Figure 2(b), querying the database with paired hashes eliminates the noise floor. The matching algorithm accurately converges on a single, definitive time offset, represented by the distinct vertical spike, confirming the structural necessity of combinatorial hashing.

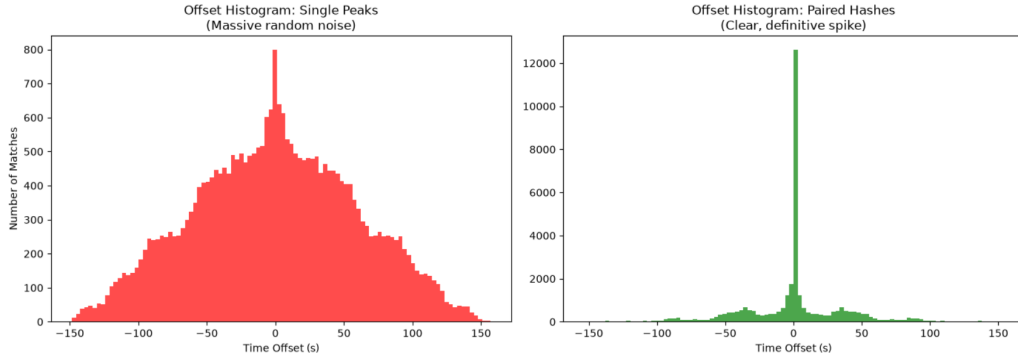


Figure 2: Histograms comparing the two matching methods. Single-peak matching (left) generates a lot of random noise and false matches, but paired hashing (right) finds the correct match, shown by the single large spike.

3 Robustness Evaluation: Noise Thresholding

To evaluate operational robustness, Additive White Gaussian Noise (AWGN) was injected into query clips at increasing multipliers. System confidence was calculated by normalizing the peak aligned hash count against the total query hashes:

$$Confidence = \frac{N_{aligned}}{N_{query}} \quad (3)$$

The injection of severe noise corrupts the feature constellation. Background noise generates random high-amplitude artifacts that trigger the 2D local maximum filter, displacing the true frequency peaks. When the true peaks are masked, the exact hash tuples $(f_1, f_2, \Delta t)$ cannot be formed, leading to lookup failures. The system demonstrates stability at lower noise multipliers but fails entirely once the magnitude of the AWGN obscures the signal's core frequency structure.

4 System Defeat: Pitch Shift and Time Stretch

Adversarial transformations applied to the query clip—specifically pitch shifting and time stretching—expose fundamental vulnerabilities in the fixed-integer hashing logic.

4.1 Failure under Pitch Shifting

A continuous pitch shift multiplies all frequencies by a constant factor. Because the Short-Time Fourier Transform utilizes discrete frequency bins, shifting the pitch pushes the spectral energy into adjacent bins. The database lookup requires absolute mathematical equality for both f_1 and

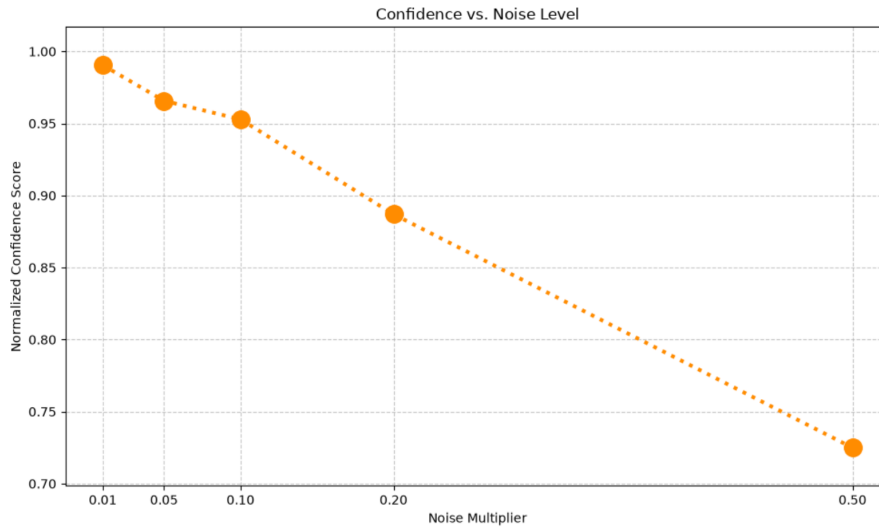


Figure 3: A graph showing how adding noise lowers the confidence score.

f_2 . Since the altered frequencies do not precisely match the indexed frequencies, the lookup returns a null result, nullifying the confidence metric.

4.2 Failure under Time Stretching

Time stretching alters the playback velocity, fundamentally changing the temporal distance between peaks. The database strictly indexes hashes based on fixed intervals Δt . When Δt expands or contracts, the 3-tuple is destroyed. While marginal correct guesses can occur due to rounding tolerances on extremely short intervals, the structural integrity of the identifier is bypassed.

4.3 Proposed Algorithmic Refinement

To render the system robust against pitch shifting, the absolute frequency values must be discarded in favor of relative metrics. Indexing the geometric ratio of the two frequencies, $\frac{f_2}{f_1}$, creates a scale-invariant hash. A constant multiplicative pitch shift applied to both frequencies cancels out, allowing the matching engine to recognize the harmonic structure regardless of the base key.

5 Conclusion

The implementation confirms that a frequency-domain audio identifier using paired constellation hashing is highly effective at isolating signals against ambient interference. However, the requirement for rigid integer matching of absolute frequencies and time intervals renders it highly susceptible to temporal and pitch-based adversarial manipulations.